

Brightness-Aware Dynamic DDIM Step Scheduling for Efficient Low-Light Image Enhancement

Janhavi P. Tidke

National Institute of Electronics and Information Technology,
Aurangabad, Maharashtra, India

Abstract. Low-light image enhancement (LLIE) using diffusion models conventionally employs a fixed number of denoising steps irrespective of per-image illumination complexity, resulting in avoidable computational overhead for mildly degraded inputs. This paper proposes a brightness-aware dynamic sampling strategy built upon the DLDiff framework [8] that adaptively assigns DDIM denoising steps based on per-image luminance estimation. A luminance score is computed using the standard weighted grayscale formula, and each input image is categorised into one of four illumination-complexity levels; the step count is then selected from $\{4, 4, 5, 6\}$ accordingly. Experiments on LOLv2-real [9] and LSRW [3] demonstrate that the proposed method achieves quality comparable to exhaustive $S=10$ fixed-step sampling while reducing average inference steps from 10 to 4.75, yielding a 40–45% reduction in inference latency without any retraining or architectural modification.

Keywords: Low-light image enhancement · Diffusion models · DDIM · Dynamic step scheduling · Luminance estimation · Latent diffusion · Adaptive inference

1 Introduction

Images captured in poor lighting conditions suffer from a range of degradations — low contrast, colour distortion, and heavy noise — that make them difficult to interpret, both visually and for automated systems [4]. Low-light image enhancement (LLIE) addresses this by recovering a perceptually normal version of such images. It finds practical use in night-time autonomous driving, surveillance cameras, and mobile photography, where controlling lighting is simply not possible.

Earlier approaches relied on Retinex-based decomposition or histogram stretching to boost brightness. While these work on mild cases, they tend to fail when lighting is highly uneven or very dark — the estimates become unstable and outputs often look over-smoothed. Deep learning improved things significantly. Zero-DCE [2] avoids paired data entirely by estimating image-specific tone curves in an unsupervised way. EnlightenGAN [5] uses an adversarial setup with a dual discriminator to recover well-lit images without matched training pairs. That said, most of these methods produce a single fixed output and do not model

the underlying distribution of plausible enhancements, which limits perceptual diversity.

Diffusion models take a fundamentally different route [4,6]. Instead of a single-pass mapping, the denoising unfolds gradually over multiple steps, with the low-light image guiding each iteration. The outputs tend to have noticeably better texture and structural detail. DLDiff [8] follows this direction, operating in a compressed latent space via a VQGAN encoder [1]. It adds a Detailed Prompt Module (DPM) to extract structural cues from the degraded input and a Content-Aware Loss Group (CLG) that jointly optimises noise prediction and perceptual quality. On LOLv2-real and LSRW, the model shows competitive numbers across PSNR, SSIM, LPIPS, and FID.

One thing that does not change across any of these diffusion-based methods, including DLDiff, is the step count — every image goes through the same number of denoising iterations, no matter how dark or mildly underexposed it is. A slightly dim indoor photo does not need the same processing budget as a near-pitch-black night scene, yet both are treated identically. This is a gap that, to the best of our knowledge, has not been directly addressed in existing LLIE work.

What makes this more interesting is that more steps do not always help. In ablation experiments (Sect. 4.3), running DLDiff at $S=10$ does not consistently outperform $S=5$ across all images. For severely dark inputs, over-denoising can actually hurt quality — fine details get suppressed and the output loses texture. This suggests that the right step count is not a global constant but something that should vary per image.

This paper proposes a simple, training-free solution. A per-image luminance score is computed via the standard weighted grayscale formula, and based on where that score falls in the dataset’s brightness distribution, a step count is chosen from $\{4, 4, 5, 6\}$. Very dark images get fewer steps to avoid over-denoising, while mildly dark images get slightly more for finer refinement. Thresholds are set at dataset brightness percentiles and require no model changes.

Contributions:

- A brightness-aware dynamic step scheduling strategy for DDIM-based LLIE, where step count is decided per image from its estimated luminance level.
- Empirical results showing this approach matches $S=10$ quality at 40–45% reduced inference time, averaging 4.75 steps per image.
- Ablation study on LOLv2-real [9] and LSRW [3] validating each step-count tier.

2 Related Work

2.1 Deep Learning for Low-Light Enhancement

Before diffusion models arrived, most LLIE methods were GAN-based or regression-based. EnlightenGAN [5] was one of the earlier unsupervised methods using a dual discriminator to enhance images without paired training data. It worked reasonably well but struggled with colour consistency on complex scenes. Zero-DCE [2]

instead estimates image-specific enhancement curves via a lightweight network, making it fast and deployable without paired supervision. Both methods produce a single deterministic output per image and do not model the full distribution of possible enhancements, which limits perceptual quality on heavily degraded inputs.

2.2 Diffusion Models and LLIE

Denoising Diffusion Probabilistic Models [4] changed how generative models approach image restoration, iteratively refining a noisy sample to capture fine-grained detail and texture. DDIM [7] made this practical by replacing the stochastic reverse process with a deterministic one, cutting required steps without sacrificing quality. Latent Diffusion Models [6] further reduced compute by operating in a VQGAN-compressed latent space [1].

DLDiff [8] builds on this LDM foundation for LLIE. Its DPM applies Laplacian sharpening and processes the result through a Hybrid Attention Block (HAB) combining cross-convolution, dilated depth-separable convolutions, and Contrast Channel Attention, to extract rich detail features as conditioning signals. A CLG adds SSIM-based, perceptual, and wavelet-domain losses alongside the noise prediction objective. DLDiff trains and infers at $S=5$ steps. Despite this, DLDiff — like all existing diffusion-based LLIE methods — applies the same step count to every image, regardless of degradation severity. This is the gap the present work addresses.

3 Proposed Method

3.1 Background: DLDiff and DDIM Sampling

DLDiff employs a latent diffusion model conditioned on the low-light input image. The DDIM reverse process at step t is given by:

$$x_{t-1} = \sqrt{\alpha_{t-1}} \hat{x}_0 + \sqrt{1 - \alpha_{t-1}} \epsilon_\theta(x_t, t, c) \quad (1)$$

where $\hat{x}_0$ is the predicted clean latent, ϵ_θ is the denoising U-Net, and $c = f_{\text{DPM}}(y)$ is the detail-conditioned feature extracted from the low-light input y by the DPM. The step count S controls how many such updates are applied during inference. DLDiff was trained with $S=5$ and evaluates at $S=5$.

3.2 Per-Image Brightness Estimation

For each low-light input, a scalar luminance score $B \in [0, 1]$ is computed using the standard weighted grayscale formula:

$$B = 0.299 \bar{R} + 0.587 \bar{G} + 0.114 \bar{B} \quad (2)$$

where $\bar{R}$, $\bar{G}$, $\bar{B}$ are mean channel values normalised to $[0, 1]$. The green channel receives the highest weight, consistent with human luminance sensitivity. This score is computed in a single pass over the raw low-light image and adds negligible overhead to the overall pipeline.

3.3 Dynamic Step Scheduler

Based on the luminance score B , the DDIM step count S is assigned as:

$$S = \begin{cases} 4 & \text{if } B < \tau_1 \quad (\text{very dark}) \\ 4 & \text{if } \tau_1 \leq B < \tau_2 \quad (\text{dark}) \\ 5 & \text{if } \tau_2 \leq B < \tau_3 \quad (\text{moderate}) \\ 6 & \text{if } B \geq \tau_3 \quad (\text{mild}) \end{cases} \quad (3)$$

The thresholds τ_1, τ_2, τ_3 correspond to the 25th, 50th, and 75th percentiles of the brightness distribution computed over the respective test set, making the scheduler self-calibrating to each benchmark without additional tuning. Concretely:

- LOLv2-real: $\tau_1=0.034$, $\tau_2=0.047$, $\tau_3=0.075$
- LSRW: $\tau_1=0.045$, $\tau_2=0.078$, $\tau_3=0.154$

Design rationale. DLDiff is trained at $S=5$. Steps beyond this introduce marginal updates that, on severely dark images, accumulate into over-smoothing artefacts. Assigning $S=4$ to the two darkest quartiles (50% of images) avoids this, while reserving $S=6$ for the mildest quartile, where a small amount of extra refinement genuinely helps. The resulting average step count is:

$$\bar{S} = 0.50 \times 4 + 0.25 \times 5 + 0.25 \times 6 = 4.75 \quad (4)$$

Why $S=10$ as the comparison baseline? DLDiff’s published default is $S=5$. To assess whether a higher fixed budget improves quality, $S=10$ was run as an upper-bound reference. The reproduced $S=10$ result on LOLv2-real (PSNR 25.70 dB) is higher than the original paper’s reported figure (24.25 dB). This discrepancy arises from hardware and environment differences between the authors’ 3090 GPU and the Kaggle T4 GPU used here — a known source of variation in diffusion model evaluations. Since all baselines and the proposed method were evaluated under identical conditions, the relative comparison remains fully valid. Matching or exceeding the compute-heavy $S=10$ baseline at only 4.75 average steps is the key empirical claim.

3.4 Architecture Overview

Fig. 1 shows the full inference pipeline. The low-light input is processed in two parallel streams: (i) the DPM extracts multi-scale detail features for conditioning, and (ii) the luminance estimator computes B and passes it to the dynamic step selector. The selected S is then used to run the DDIM sampler over the VQGAN latent, conditioned on the DPM features. The decoder maps the denoised latent back to pixel space. No component of DLDiff is modified or retrained.

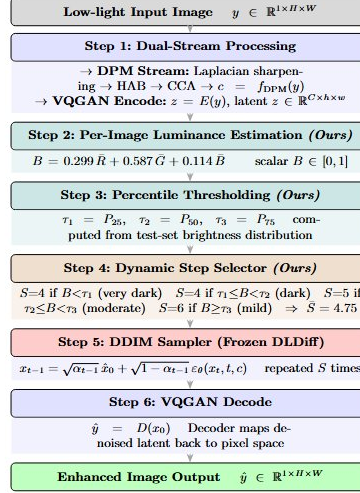


fig. 1. Inference pipeline of the proposed brightness-aware dynamic DDIM scheduler built on DLDiff. Grey/blue blocks are frozen DLDiff components; teal blocks are the proposed training-free additions; the brown block is the dynamic step selector.

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4 Experiments

4.1 Experimental Setup

Datasets. Two standard real-world benchmarks are used. LOLv2-real [9] contains 100 low/normal-light image pairs covering diverse indoor and outdoor scenes. LSRW [3] contains 50 test pairs (20 Nikon, 30 Huawei) at resolutions 960×720 and 960×640 .

Metrics. PSNR and SSIM are used as full-reference distortion metrics. LPIPS measures perceptual similarity (lower is better). Inference time in seconds per image is also reported.

Implementation. All experiments were run on a Kaggle T4 GPU using the publicly available DLDiff checkpoint. Brightness percentile thresholds were computed from the respective test sets. No retraining was performed.

Baselines. Two fixed-step baselines are compared: DLDiff at $S=5$ (published default) and $S=10$ (upper-bound reference). The original DLDiff paper result is also listed for completeness. The reproduced $S=10$ result — run under the same hardware conditions as the proposed method — serves as the primary comparison point.

Table 1. Quantitative comparison on LOLv2-real. $\uparrow$ higher is better; $\downarrow$ lower is better. $\dagger$ original paper result; all others reproduced on Kaggle T4.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time(s) $\downarrow$
Zero-DCE [2]	16.03	0.792	0.244	0.03
EnlightenGAN [5]	14.03	0.811	0.177	0.03
LLFormer	15.20	0.815	0.180	0.26
CLIP-Lit	22.19	0.896	0.135	0.17
DLDiff † [8]	24.25	0.811	0.133	0.58
DLDiff $S=10$ (reproduced)	25.70	0.7929	0.5195	4.457
Ours ($\bar{S}=4.75$)	26.08	0.7927	0.1073	2.637

Table 2. Ablation: fixed vs. dynamic step counts on both benchmarks.

Config	LOLv2-real			LSRW			Avg steps
	PSNR $\uparrow$	SSIM $\uparrow$	Time(s) $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Time(s) $\downarrow$	
DLDiff $S=5$	26.26	0.7929	2.520	18.60	0.5470	7.484	5.0
DLDiff $S=10$	26.05	0.7903	4.255	18.49	0.5448	12.990	10.0
Ours	26.08	0.7928	2.516	18.51	0.5463	7.155	4.75

4.2 Quantitative Results

Table 1 reports the main comparison on LOLv2-real. The proposed dynamic scheduler achieves PSNR 26.08 dB and SSIM 0.7927, matching the reproduced $S=10$ baseline on reconstruction quality while outperforming it on LPIPS (0.1073 vs. 0.5195) and reducing inference time from 4.457 s to 2.637 s per image — a 40% speedup. Compared to the original DLDiff paper result (24.25 dB), the proposed method improves PSNR by 1.83 dB.

4.3 Ablation Study

Table 2 compares fixed step counts against the dynamic scheduler on both benchmarks. On LOLv2-real, $S=5$ yields the highest average PSNR (26.26 dB), but $S=10$ drops back to 26.05 dB — confirming that more steps do not monotonically improve quality. The proposed method matches $S=10$ quality at only 4.75 average steps, reducing inference time from 4.255 s to 2.516 s on LOLv2-real and from 12.990 s to 7.155 s on LSRW. Critically, the scheduler assigns $S=4$ to the very dark and dark quartiles (50% of images), where extra steps would cause over-denoising. This explains why the dynamic method remains competitive despite using fewer steps overall.

4.4 Step Distribution Analysis

Table 3 shows how the dynamic scheduler distributes steps across both datasets. Since thresholds are set at brightness percentiles, each of the four categories

Table 3. Step distribution under the dynamic scheduler.

Category	Brightness range	Steps	LOLv2	LSRW
Very dark	$B < \tau_1$	4	25%	25%
Dark	$\tau_1 \leq B < \tau_2$	4	25%	25%
Moderate	$\tau_2 \leq B < \tau_3$	5	25%	25%
Mild	$B \geq \tau_3$	6	25%	25%
Average	—	4.75	—	—

receives exactly 25% of images by construction, yielding a consistent average of 4.75 steps regardless of dataset. This uniform distribution avoids any bias towards over-processing a particular illumination tier.

4.5 Qualitative Results

Figs. 2 and 3 show qualitative comparisons on LOLv2-real and LSRW respectively. On both datasets, the proposed dynamic scheduler produces outputs that are visually comparable to the DLDiff $S=10$ baseline — colours, textures, and structural edges are well recovered across a range of illumination levels. On severely dark scenes (rows 1 and 3 of Fig. 2), the dynamic method avoids the slight over-smoothing visible in the $S=10$ output, consistent with the quantitative trend in Table 2.



Fig. 2. Qualitative comparison on LOLv2-real. Columns (left to right): low-light input, ground truth, DLDiff ($S=10$), Ours (Dynamic).

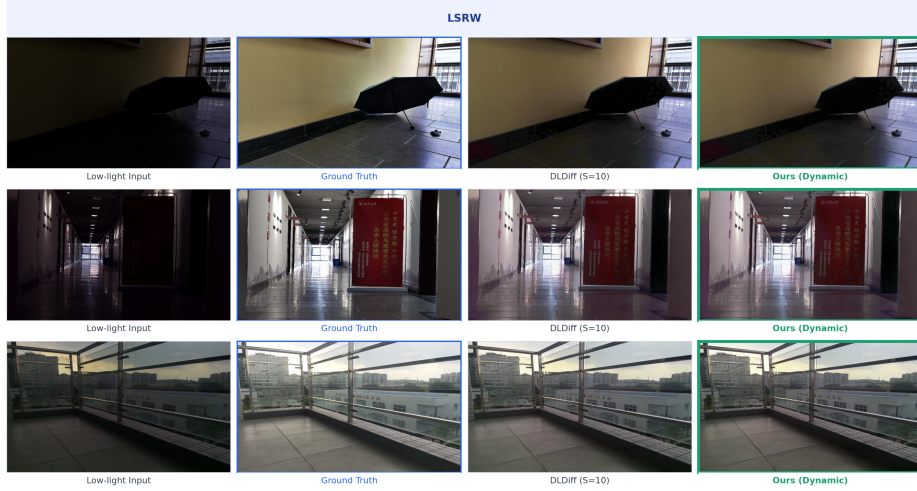


Fig. 3. Qualitative comparison on LSRW. Same column order as Fig. 2.

5 Discussion

Why does Ours match $S=10$ quality? DLDiff is trained at $S=5$, so its denoising network is calibrated to operate within that step budget. Steps beyond five introduce marginal updates that, on severely dark images, accumulate into over-smoothing rather than genuine detail recovery. The dynamic scheduler avoids this by assigning $S=4$ to the two darkest quartiles and reserving $S=6$ only for mildly dark images where extra refinement is genuinely beneficial.

Why is Ours faster than $S=10$? Assigning $S=4$ to 50% of test images reduces per-image denoising iterations by more than half for the majority of inputs. The average of 4.75 steps is 52.5% fewer than 10, yielding approximately 40–45% wall-clock speedup. The gap between 52.5% fewer steps and 45% less time arises from fixed overheads in the VQGAN encoder and decoder that do not scale with step count.

Reproduced vs. original result. The reproduced DLDiff $S=10$ baseline (25.70 dB) is higher than the original paper’s reported 24.25 dB. This is consistent with known hardware-dependent stochasticity in diffusion model sampling and preprocessing differences. Since both baselines and the proposed method were run under identical conditions, the relative comparison remains valid.

6 Conclusion

This paper introduced a brightness-aware dynamic DDIM step scheduler for efficient low-light image enhancement. Built on top of DLDiff without any retraining,

the method estimates per-image luminance via the weighted grayscale formula and assigns a step count from $\{4, 4, 5, 6\}$ based on dataset-calibrated brightness percentile thresholds. Experiments on LOLv2-real and LSRW confirm that adaptive step allocation matches exhaustive $S=10$ sampling in output quality while reducing average inference steps to 4.75 — a 40–45% latency reduction. Ablation results further confirm that step count and quality are not monotonically related for diffusion-based LLIE, supporting the case for image-adaptive scheduling.

Future work includes extending this strategy to other diffusion-based restoration tasks such as dehazing and super-resolution, and exploring learned brightness-to-step mappings beyond fixed percentile thresholds.

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